Tau Core: An Atemporal Morphological Readout Framework and a Conditional Minimum-Viable Physical Completion

Foundation Paper I

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Abstract

Tau Core asks whether observed four-dimensional physics can be represented as an observer-indexed, information-losing readout of a richer atemporal morphological object. We define the minimal architecture $s \mapsto \rho_\tau(s) \mapsto M_\tau \mapsto \mathrm{Desc}_O(M_\tau)$, enforce source-freeze, null-quotient and no-double-counting rules, and fix one minimum-viable completion. There a stabilized body is solved before terminal variation, while one once-counted functional supplies gravitational, gauge/current, quantum-effect and clock rows. Under explicit Lorentzian, regularity, state/effect and observer-carrier assumptions, this construction has compatible conditional limits: a semiclassical Einstein equation with renormalized quantum stress and the fixed-background Schrödinger–Born description.

The same completion yields the single-record resolution bound

$$\delta t_{\min,O} \geq \frac{2\hbar}{\Delta H} \arcsin \frac{\epsilon_O}{2} \geq \frac{\hbar \epsilon_O}{\Delta H},$$

where $x_A = \epsilon_O^2$ is an independently reconstructed dimensionless body-action separation. Uhlmann minimization in the declared common-Gram completion gives $x_A = x_Q = 2(1 - \sqrt{F})$ and $D_B = 2 \arcsin(\epsilon_O/2)$. The proposed Tau-specific discriminator is this independently testable body–Uhlmann identity, not the standard quantum speed limit or its inverse-energy dependence.

The main source result is conditional but finite. Inside an enriched special dagger-Frobenius record completion, a complete source-owned record edge induces a faithful trace-preserving standard-form map; primitive-defect extensivity then forces the exact Trace–Gram identity on its endpoint pair and composable path closure. Endpoint-only descent is permitted only when relative holonomy is fixed in advance as Uhlmann-gauge or null under the complete endpoint descriptor. This is a sufficient internal realization theorem, not a derivation that complete edges exist physically. Record occupation, the finite identity and operational hold are independent required parts of a test. Nature-level source selection, laboratory occupation, arbitrary-state coverage, numerical constants, ultraviolet/Standard-Model completion and empirical confirmation remain open. The paper therefore presents an auditable conditional completion, not a replacement for general relativity or quantum theory.

Manuscript Roadmap

1 Purpose and claim boundary

This manuscript states the Tau Core foundation language and the strongest currently frozen physical completion in a form that can be criticized, reproduced, and rejected. It is self-contained: no result from another Tau Core manuscript, repository, or unpublished analysis is used as a premise or as evidence for any claim made here.

The claim boundary has two levels. At the foundational level, the paper defines an atemporal body/readout architecture and its audit discipline. At the completion level, it fixes one common functional and proves consequences conditional on an explicit assumption ledger. Neither level supplies empirical validation or a theorem that the physical base–seed source selects that completion. The main exclusions are summarized below and itemized again near the end of the manuscript.

Reviewer-facing claim box. This paper claims (i) a formal body/readout architecture and failure discipline, and (ii) one *conditional* minimum-viable completion with compatible GR/optics and quantum/Born limits plus one frozen observer-resolution lower-bound discriminator. It does not claim empirical validation, uniqueness, nature-level source selection, a new force, dark-matter or dark-energy replacement, ultraviolet quantum gravity, or full Standard-Model recovery. Every conditional result below is void as a claim about Nature unless the common completion is independently selected or its predeclared discriminator survives testing.

The central idea can be written in one line:

$$s \longmapsto \rho_\tau(s) \longmapsto U_i^{M_\tau}(\rho_\tau(s)). \tag{1}$$

The first arrow is not time evolution. It is a parent-side response relation. The second arrow is a body-conditioned readout. The "readout" is not an extra physical substance or history store: in the completion developed below it is an observer-specific realization of ordering and compatibility already resident in the stabilized body. A physical branch appears only after the descent has supplied a codomain, a null policy, a closure test, and a validation boundary. When M_τ is fixed by the declared source class, the superscript is often suppressed and U_i is written for $U_i^{M_\tau}$.

In a deterministic branch one may write

$$\rho_\tau : \mathcal{S} \to \mathfrak{R}_\tau.$$

In a relational branch the same symbol denotes either a relation

$$\rho_\tau \subseteq \mathcal{S} \times \mathfrak{R}_\tau$$

or, equivalently, a set-valued response rule

$$\rho_\tau : \mathcal{S} \to \mathcal{P}(\mathfrak{R}_\tau).$$

Such a branch must declare whether it uses the whole admissible response class or an endpoint-blind source-freeze rule selecting a representative.

Equivalently, the paper works with the following foundation package:

$$\mathfrak{T} = (\mathcal{S}, \mathfrak{R}_\tau, \rho_\tau, M_\tau, \{U_i\}, \{N_i\}, \{J_i\}).$$

Here $\mathcal{S}$ is a seed/source domain, $\mathfrak{R}_\tau$ is a formal response domain, ρ_τ is a map, relation, or set-valued response rule, M_τ is the morphological body, U_i are sector readouts, N_i are null or quotient policies, and J_i are optional relifts. This package is an organizing object for later branches (Fig. 1), not a proof that Nature uses it.

In informal Tau Core discussions, the phrase *Tau Core base field* refers to this parent-side response substrate, represented here by $(\mathcal{S}, \mathfrak{R}_\tau, \rho_\tau)$, together with the morphological body M_τ. The phrase is heuristic. In this paper it does not denote a physical field with an action, equations of motion, units, stress-energy, or empirical calibration.

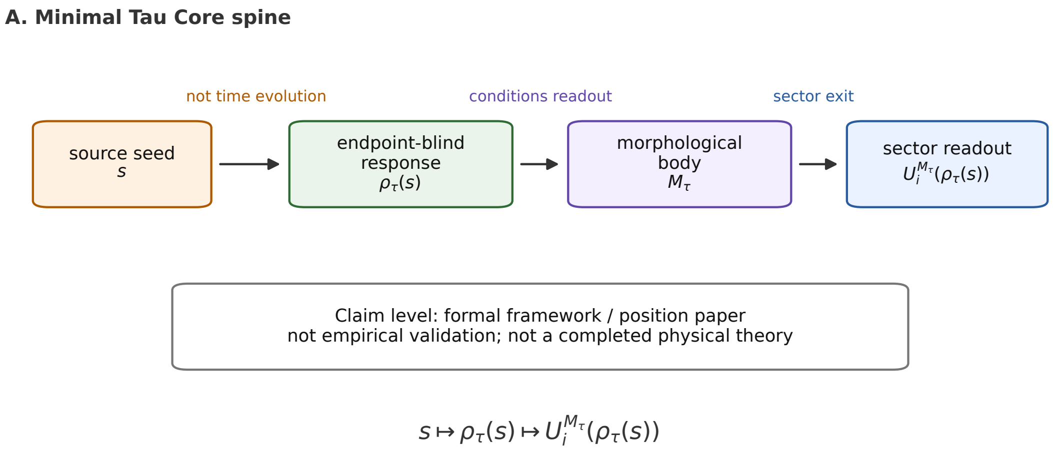

Figure 1: The minimal Tau Core spine used in Foundation Paper I. A parent-side seed s induces an endpoint-blind Tau response $\rho_\tau(s)$. A Tau-side morphological body M_τ conditions sector readouts U_i. The diagram is the weak foundation spine; the stronger conditional completion is shown in Fig. 4.

Method note. The manuscript was developed in an AI-assisted theoretical-research workflow. The AI workflow is not used as evidence for any physics claim; it is only part of the drafting and audit process.

This paper is therefore neither a phenomenology paper nor only a position paper. It is a foundation paper containing a conditional worked completion. It is useful only if that completion makes the program more falsifiable. A broad ontology that cannot be turned into frozen source rules, null tests, or endpoint-blind predictions is not useful as physics.

2 Foundation architecture and formal discipline

The paper has three jobs: define the body/readout architecture, exhibit one conditional physical completion, and state the exact physical-selection and empirical boundaries. The motivating question is whether time, spacetime, quantum state and other terminal descriptions can be observer-indexed readouts of a deeper response rather than mutually independent primitives. Related timeless, relational-time, reconstruction and constructor-theoretic programs show why this question is legitimate, but none is imported as evidence for Tau Core [9, 18, 7, 12, 6, 8].

The historical symbol τ names the parent side, not an additional parent clock. A stable four-dimensional block is therefore typed as

$$M_{4D} = U_{4D}^{M_\tau}(\rho_\tau(s)), \tag{2}$$

rather than assumed to be the final ontology. Ordinary spacetime field theory remains the effective theory on that descended leaf. This distinction has physical content only when another source-frozen readout imposes a constraint that is not already contained in the 4D endpoint.

The primitive vocabulary is finite. A seed $s \in \mathcal{S}$ is selected without endpoint leakage. The endpoint-blind response ρ_τ may be a map, relation or set-valued rule. Candidate realizations include a variational extremum, constraint solution, algebraic closure, probabilistic kernel, fixed point, or functorial construction; this paper does not identify those mathematically different mechanisms. The morphological body M_τ is the intrinsic structured object conditioning the readouts, not an observed 4D morphology label. A readout is a typed map

$$U_i : \mathcal{M}_\tau^{\mathrm{adm}} \times \mathfrak{R}_\tau \to \mathcal{O}_i$$

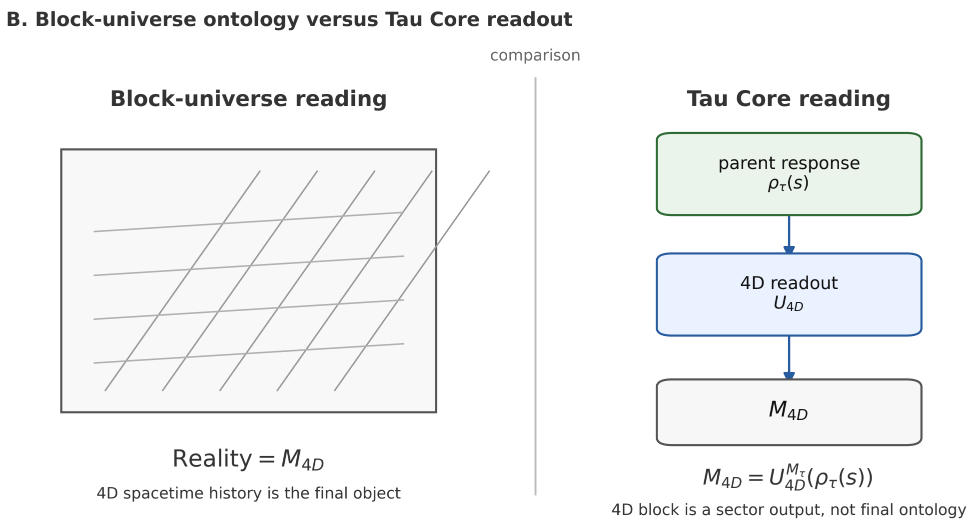

Figure 2: A four-dimensional block may be an effective final description for one readout while remaining a descended sector of the Tau parent. This is a positioning distinction, not a refutation of block-universe accounts.

with a declared codomain, null policy and validation boundary. A relift $J_i : \mathcal{O}_i \dashrightarrow \mathfrak{R}_\tau$ is optional and need not be unique. If it is supplied, the coordinate-free obstruction is

$$\omega_i(r) = 0 \quad \Longleftrightarrow \quad J_i U_i(r)\, N_i\, r, \tag{3}$$

while $J_i U_i(r) - r$ is only a chart-level representative. These definitions do not themselves construct the physical body or explain a dark sector.

Every readout induces the equivalence relation

$$r \sim_i r' \quad \Longleftrightarrow \quad U_i(r) = U_i(r'). \tag{4}$$

A joint readout has equivalence relation $\sim_i \cap \sim_j$ and therefore refines each component. Refinement is not statistical independence: several readouts may transmit the same parent degree of freedom. Tau Core accordingly requires a source-overlap ledger and forbids counting source-overlapping terminal terms as independent corrections unless their split or covariance is fixed upstream.

The weak-spine status is correspondingly strict: $\mathcal{S}$, $\mathfrak{R}_\tau$, ρ_τ, M_τ, U_i, N_i, J_i and ω_i are typed definitions or declared domains. The MVP below adds one stationary body, positive quotient Hessian and gravity/optics, quantum/Born and QOR leaves, but does not prove that Nature selects or occupies them.

The terminal inventory is likewise explicit rather than unlimited. The spacetime/gravity family must recover metric, causal and lensing structure; observer-time must specify ordering, clock normalization and observer dependence; quantum access must recover state, effect, Born and no-signalling structure; matter/source must identify occupied modes and stress-energy; and the cosmological family must recover expansion, redshift and early-universe limits. Electromagnetic, gauge and composite astrophysical observables are later typed descendants of this common packet, not permission to introduce an independent correction for every anomaly. None of these names counts as a separate information source until overlap and factorization have been proved.

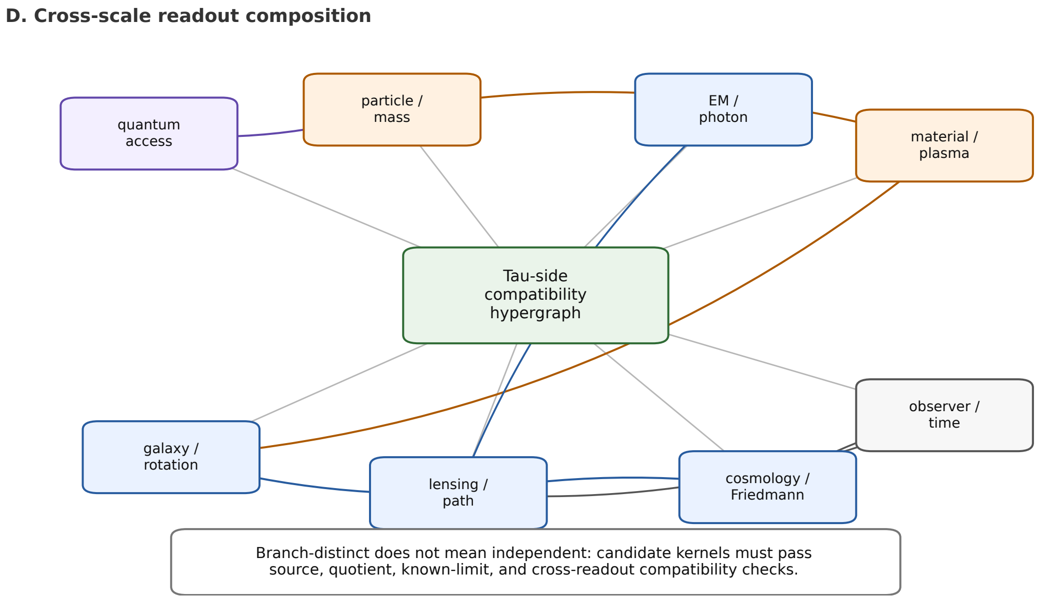

Figure 3: Readout families are branch-distinct but not presumed independent. Source, carrier, quotient and compatibility relations must be audited before separate terminal names are counted as separate information.

3 A conditional minimum-viable physical completion

The preceding architecture is deliberately weak: by itself it does not select a physical action, metric signature, quantum state, or observer. This section fixes one minimum-viable physical (MVP) object so that the paper contains an auditable theory chain rather than an indefinitely extensible list of possible readouts. The frozen chain includes a quantum observer-resolution (QOR) discriminator and is

$$\boxed{\begin{aligned}\Gamma_{\mathrm{MVP}} \longrightarrow M_* \longrightarrow \mathrm{Desc}_O(M_*)\\ \longrightarrow \{\mathrm{GR/optics}, \mathrm{quantum/Born}, \mathrm{QOR}\}.\end{aligned}} \tag{5}$$

Every arrow in Eq. (5) is conditional. The chain is internally composable and has a frozen discriminator, but upstream selection in Nature remains open (Fig. 4).

3.1 Frozen assumption ledger

The completion uses the following assumptions. They are deliberately separated from consequences.

Label	Assumption used in the MVP completion
A1	A physically enriched base and one universal seed admit a stabilized body M_* with a twice-differentiable body action.
A1-SHR	On the primary finite SHR branch, the seed owns a separating Alexandrov possibility family with at least one strict prerequisite cover; seed and body entries related by the realization null are two presentations of one pure deformation.
A1-USC	The universal seed owns every primitive ordering distinction of the occupied body response. A source-owned decoder $q\widehat{R} = I$ maps the complete stabilized body realization back to that seed order; a terminal-visible decoder-null ordering component would falsify this assumption.

Label	Assumption used in the MVP completion (continued)
A2	After quotienting only directions annihilated by every claimed terminal, the body Hessian K_M is positive on the retained tangent space.
A3	The body is solved first and frozen; terminal variations do not re-stabilize it or feed endpoint information into its definition.
A4	Coframe, connection/gauge, occupied matter, clock, and effect probes are once-counted terms or sources of one post-body generating functional.
A4-SRCMIN	The minimum-viable source ledger admits no independently occupied primitive carrier unless it has a separately typed, rank-novel recovery role. In particular, no independent base causal-work carrier is present in the MVP.
A5	The relevant descended leaf is orientable and Lorentzian; its gravitational term lies in the torsion-free Palatini/Einstein–Hilbert infrared class and is sufficiently regular for the variations used below.
A6	The quantum leaf admits positive normalized states and a complete effect family; the valuation on complete effects is additive and noncontextual.
A7	The field-theory leaf admits a locally covariant Hadamard prescription for the renormalized stress tensor and its local curvature ambiguities.
A8a-OCC	The physical completion occupies a nonzero source-owned recovery support, at least one record incidence, and a CP record instrument on that support. The finite pointer algebra and common Uhlmann amplitude bundle are then derived, not separately assumed.
A8b	The same post-body source that owns $x_O = \mathcal{A}_X^{PC}/\mathcal{A}_*$ selects a normalized amplitude connector whose horizontal Hilbert–Schmidt pullback equals the normalized body-action metric. On the explicit mixed-state completion it selects $V_{\text{phys}} = V_{r(x_O)}$, with $r(x_O)^2 = x_O - x_O^2/4$ and $0 < x_O \leq 2$. In the enriched complete-record-edge completion, the connector and its finite identity are derived from the common trace/action character; physical occupation of that completion remains open.
A9	The quantum leaf uses the common action unit $\mathcal{A}_* = \hbar$, and self-adjoint transport obeys the Fubini–Study/Bures quantum-speed-limit bound.

A1–A4-SRCMIN, including A1-USC, define the working completion. A5–A9, including A8a-OCC, are standard-limit and prediction assumptions. None is claimed to follow from the old minimal Tau skeleton alone. In particular, positivity of a body Hessian does not select the Lorentzian leaf, the Born valuation, or a physical observer carrier. Section 3.9 proves a sufficient algebraic source theorem for A8b and then derives affiliation inside the declared primary SHR minimum-source completion. A8b is therefore a conditional theorem of that MVP packet, not a theorem of old-bare Tau or a nature-level selection result.

The physical motivation for A1-USC is source economy plus order consistency, not mathematical necessity. A universal seed is intended to be the compact owner of the primitive possibility and prerequisite distinctions that the base realizes redundantly as a body. If the realization produced an additional primitive order with no seed preimage, then two bodies agreeing on the universal seed could disagree on causal incidence; the seed would no longer be universal in the stated sense. A1-USC excludes that ambiguity and makes the body a realization of one source order. This motivation does not prove that Nature adopts such a seed: an order-generating base interaction is an admitted competing completion and is an explicit falsifier of A1-USC.

3.2 Body-first variational order

Let $\mathcal{A}_{\text{body}}$ denote the atemporal body action. Stabilization means

$$D_M \mathcal{A}_{\text{body}}(M_*) = 0, \qquad K_M = D_M^2 \mathcal{A}_{\text{body}}\big|_{M_*} > 0 \quad \text{on } T_{M_*}\mathcal{M}/\mathcal{N}_{\text{true}}. \tag{6}$$

Here $\mathcal{N}_{\text{true}}$ contains only directions annihilated by every terminal retained in the MVP. A direction invisible to one terminal is not discarded if another terminal sees it.

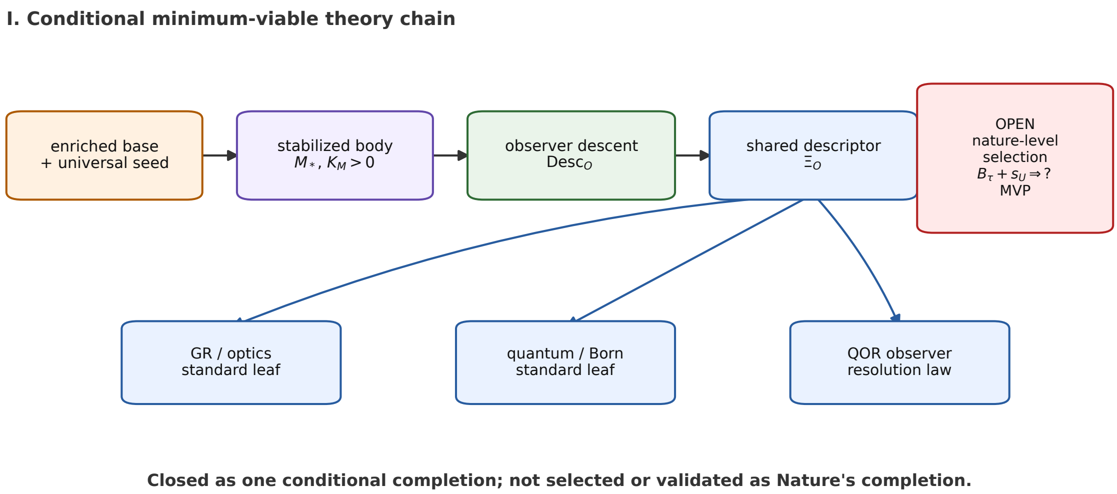

Figure 4: The conditional minimum-viable chain. A physically enriched base and universal seed are assumed to stabilize a body before observer descent. One shared descriptor then supplies the standard leaves and the QOR discriminator. The red box is the exact open step: nature-level selection of this completion from the upstream base–seed source.

The order in Eq. (6) matters. If z denotes all post-body terminal variables, linear response is solved in the ordered form

$$K_M\,\delta M = -j_M, \tag{7}$$

$$K_z\,\delta z = -j_z - C\,\delta M, \tag{8}$$

where $C = D_z D_M \Gamma|_{M_*}$ is an allowed body-to-terminal coupling. Equations (8) define a lower-triangular response problem. Placing M and z in an unconstrained simultaneous symmetric fit would reverse the generative direction and permit endpoint data to alter the body. The completion therefore forbids that operation unless a separately sourced post-body copy of the body variable is introduced and justified.

3.3 One common post-body functional

Three differently typed objects must be kept separate: the pre-body action used to solve M_*, the post-body microscopic functional before quantum or apparatus variables are integrated out, and the renormalized observer-indexed effective functional afterwards. Their lawful order is

$$\mathcal{A}_{\rm pre}[M; s, B] \xrightarrow{\mathrm{stat}_M} M_*, \qquad \Gamma_{\rm micro,O}[\Phi; M_*, e, A, \eta], \qquad \Gamma_{\rm MVP} \equiv \Gamma_{\rm eff,O}[M_*; e, A, \eta]. \tag{9}$$

The transferable body sector used in the first solve is not appended again downstream. An admissible once-counted microscopic ownership ledger is

$$\begin{aligned}\Gamma_{\rm micro,O} = {} & S^{\rm bare}_{\rm grav}[e] + S^{\rm ret}_{\rm body}[M_*; e, \Phi_{\rm ret}] \\ & + S^{\rm bare}_{q}[M_*; e, A, \Phi_q] + S_{\rm src,O}[M_*; e, A, \Phi_q, \Phi_{\rm ret}; \eta, \vartheta],\end{aligned} \tag{10}$$

where every primitive monomial or incidence has exactly one owner. Integrating the declared quantum and apparatus variables once defines

$$e^{iW^{\rm ren}_O[M_*; e, A, \Phi_{\rm ret}, \eta, \vartheta]/\hbar} = \int_{\mathcal{C}_O} \mathcal{D}\Phi_q\, e^{i(S^{\rm bare}_q + S_{\rm src,O})/\hbar}\Big|_{\rm ren} \tag{11}$$

and the effective packet

$$\boxed{\Gamma_{\mathrm{MVP}} = S_{\mathrm{EH}}[e] + S^{\mathrm{ren}}_{\mathrm{loc}}[e] + S^{\mathrm{ret}}_{\mathrm{body}}[M_*; e, \Phi_{\mathrm{ret}}] + W^{\mathrm{ren}}_{O}[M_*; e, A, \Phi_{\mathrm{ret}}, \eta, \vartheta].} \tag{12}$$

Here $S^{\mathrm{ret}}_{\mathrm{body}}$ is the retained post-body representative; its role is to preserve the occupied body value and first variation without copying the already used pre-body envelope. If observer degrees of freedom and their source insertions are already included in W^{ren}_{O}, there is no additional Γ_{obs} summand at the effective level. A separate observer term is admissible only when it owns a disjoint apparatus field domain.

This gives an exact de-duplication rule at quadratic order. For a Hessian block

$$K = \begin{pmatrix} K_x & B \\ B^\dagger & K_y \end{pmatrix}, \qquad K_{\mathrm{eff}} = K_x - B K_y^{-1} B^\dagger, \tag{13}$$

one may retain the full block or eliminate y and retain its Schur reduction, but not both. Propagators and Green response operators are inverses of second variations of Eq. (12); they are not additional action or stress summands. Likewise, for any allowed finite local covariant counterterm $C_{\mathrm{loc}}[e]$, the relocation

$$S^{\mathrm{ren}}_{\mathrm{loc}} \mapsto S^{\mathrm{ren}}_{\mathrm{loc}} + C_{\mathrm{loc}}, \qquad W^{\mathrm{ren}}_{O} \mapsto W^{\mathrm{ren}}_{O} - C_{\mathrm{loc}} \tag{14}$$

leaves Γ_{MVP} and its full variations unchanged. Adding the same counterterm to both terms would change the equation and is double counting. Equations (10)–(14) prove a once-counted effective and quadratic ledger once a microscopic ownership partition and renormalization prescription are supplied. They do not construct the full nonlinear microscopic Tau action or its ultraviolet completion.

For an occupied observer carrier O, define the typed shared descriptor

$$\Xi_O = \mathrm{Desc}_O\left(M_*, K_M, e, A, \psi, \vartheta, \{F_{O,a}\}\right). \tag{15}$$

More formally, on the adopted leaf

$$\mathrm{Desc}_O : \mathcal{D}_{M_*}/\mathcal{N}_{\mathrm{true}} \longrightarrow \mathcal{X}_O,$$

where $\mathcal{D}_{M_*}$ is the typed post-body data bundle and $\mathcal{X}_O$ is the shared observer descriptor space. Quotient safety requires $\mathrm{Desc}_O(x) = \mathrm{Desc}_O(x')$ whenever $x \sim_{\mathrm{true}} x'$ under the retained common-null relation. The descent is the observer-specific realization of the ordering and compatibility law resident in the body. It is not a separate material layer, does not store a trajectory, and is not defined as a correction to a standard model. Standard physics and any future nonstandard component belong to the same full descent.

The terminal rows are generated by variations of the same packet:

$$\mathcal{E}_e = \frac{\delta \Gamma_{\mathrm{MVP}}}{\delta e}, \qquad \mathcal{E}_A = \frac{\delta \Gamma_{\mathrm{MVP}}}{\delta A}, \tag{16}$$

$$p(a|O, M_*) = \left.\frac{\delta W_O}{\delta \eta_a}\right|_{\eta=0}, \qquad \mathcal{E}_\vartheta = \frac{\delta \Gamma_{\mathrm{MVP}}}{\delta \vartheta}. \tag{17}$$

Common ownership fixes cross-sector covariance and prevents duplicate matter or Green-response terms (Fig. 5).

3.4 Conditional gravity and optical leaf

On A1–A5 and A7, the low-curvature gravitational part of the completion has the effective form

$$\Gamma_g = \frac{1}{16\pi G_{\mathrm{SHR}}} \int d^4x \sqrt{-g}\,(R - 2\Lambda_{\mathrm{ren}}) + S^{\mathrm{ren}}_{\mathrm{loc}}[g] + W^{\mathrm{ren}}_{O}[g, A, \Phi_{\mathrm{ret}}, \eta, \vartheta; M_*], \tag{18}$$

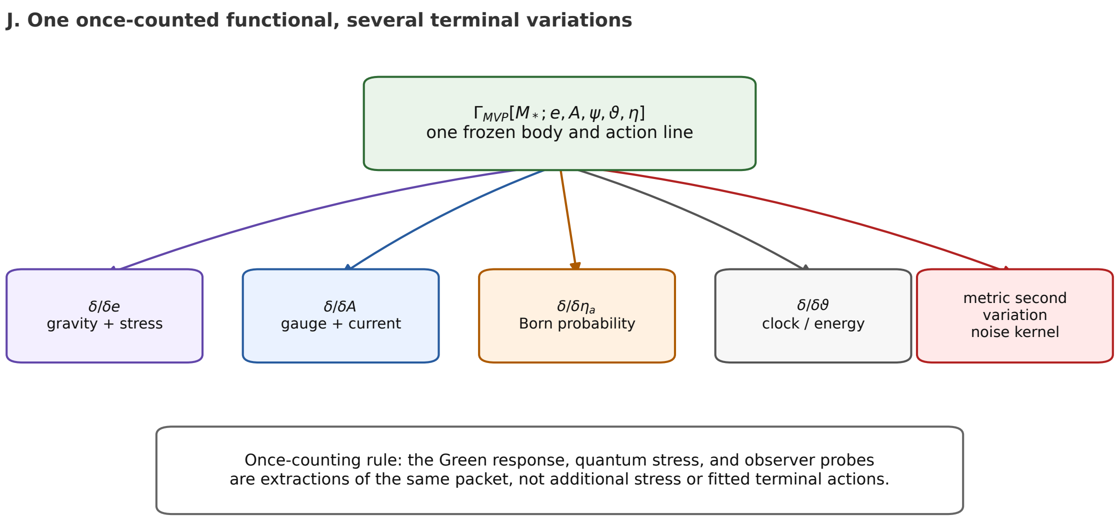

Figure 5: Terminal variations of one once-counted post-body functional. The diagram is a dependency map, not a claim that every displayed sector is already selected by the physical Tau parent.

where $S_{\rm loc}^{\rm ren}$ collects allowed local curvature counterterms and $W_O^{\rm ren}$ is the once-integrated quantum/observer functional. Coframe or metric stationarity gives

$$G_{\mu\nu} + \Lambda_{\rm ren} g_{\mu\nu} + H_{\mu\nu}^{\rm loc,ren} = 8\pi G_{\rm SHR} \left(T_{\mu\nu}^{\rm body,ret} + \langle T_{\mu\nu}^{q+O} \rangle_{\rm ren} \right). \tag{19}$$

Here $T_{\mu\nu}^{\rm body,ret}$ is the metric derivative of the retained body replacement term and excludes every degree of freedom integrated into $W_O^{\rm ren}$. The second term is the metric derivative of that same effective functional and is not added as a separate microscopic action. Local covariance and diffeomorphism invariance give the corresponding Ward identity and conservation of the *complete* right-hand-side packet. The separate body and quantum pieces need not be conserved before their coupling and renormalization prescription are fixed. The structure of Eq. (19) is the standard semiclassical-gravity structure, including finite local curvature ambiguities [21, 13, 14, 15]; the Tau claim is its conditional common-source placement, not novelty of the equation.

If quantum stress, stress fluctuations, and local higher-curvature terms are negligible, Eq. (19) reduces to the Einstein equation. Its weak-field, slow-motion leaf gives the Poisson equation and Newtonian motion. For a source-free Abelian radiation field, the same Lorentzian leaf gives the eikonal conditions

$$k^\mu k_\mu = 0, \qquad k^\nu \nabla_\nu k^\mu = 0, \tag{20}$$

and the usual redshift and optical Jacobi equations. No independent lensing gain is introduced. These are conditional standard-limit recoveries; the completion does not predict numerical values of $G_{\rm SHR}$ or $\Lambda_{\rm ren}$.

3.5 Conditional quantum and Born leaf

At fixed descended metric, let ρ_O be a positive normalized state and $\{F_{O,a}\}$ a positive complete effect family,

$$\rho_O \geq 0, \quad \mathrm{Tr}\,\rho_O = 1, \qquad F_{O,a} \geq 0, \quad \sum_a F_{O,a} = I. \tag{21}$$

Under A6, complete additive noncontextual valuation yields the trace form

$$\boxed{p(a|O, M_*) = \mathrm{Tr}(\rho_O F_{O,a}).} \tag{22}$$

This is a Gleason-type representation result [10, 5]. Its precise domain assumptions matter: ordinary projective Gleason theory requires Hilbert dimension at least three, while generalized-effect versions use stronger measurement richness to cover the two-dimensional case. Tau Core does not infer Eq. (22) from information loss alone; positivity, completeness, and valuation structure are indispensable assumptions of the frozen completion.

At fixed background and in a regular positive-frequency sector, the same functional gives the Schrödinger description and locally covariant quantum field theory. A Hadamard prescription supplies $\langle T^{q+O}_{\mu\nu}\rangle_{\mathrm{ren}}$ in Eq. (19). Microcausality together with local completely positive trace-preserving operations gives operational no-signalling; entanglement changes joint statistics without creating a controllable superluminal signal. These are recovery conditions shared with standard algebraic QFT in curved spacetime [4, 13, 14], not novel predictions of the Tau architecture.

3.6 Compatibility of the standard limits

The gravity and quantum leaves are not two separately fitted models. They are limits of Eq. (12). Holding the descended metric fixed gives the quantum/Born leaf. Varying the coframe while retaining the renormalized quantum expectation gives Eq. (19). Suppressing quantum stress fluctuations and then taking the classical matter limit reaches the same classical geometry as taking the classical limit first and varying the corresponding classical matter action.

Proposition 1 (Conditional compatibility of the standard leaves)**.** *Under A1–A7, a common renormalization convention, and the stated regularity and limit–variation interchange assumptions, the fixed-background quantum limit and the small-fluctuation Einstein limit are compatible at the displayed semiclassical order. In the joint limit $\hbar \to 0$, suppressed stress noise, and negligible local higher-curvature terms, both routes reduce to classical matter on the same Einstein geometry.*

Conditional argument. Both routes are first variations of the same once-counted functional. The metric derivative of the renormalized matter term is the quantum stress in Eq. (19); replacing the state by its classical limit turns that derivative into classical matter stress. Because the body and renormalization convention are held fixed, no independent coefficient is introduced when the order of the two limits is exchanged. The stated regularity assumptions justify interchange of the limit and first variation. □

This is a completion-internal consistency result, not a proof of nonperturbative quantum gravity, metric quantization, or uniqueness (Fig. 6).

3.7 QOR: a frozen observer-resolution discriminator

The MVP does not derive Lorentzian gravity or quantum state/effect structure from the weak Tau spine; it tests whether these adopted structures can be jointly organized as terminals of one frozen body-first completion. To do more than rewrite known equations, A8a-OCC together with A8b provides two source-frozen stable record prototypes $r_{O,0}$ and $r_{O,1}$ in the occupied observer carrier $\mathcal{C}_O$. Let $\iota_O : \mathcal{C}_O/N_O \to T_{M_*}\mathcal{M}/\mathcal{N}_{\mathrm{true}}$ be its source-declared linearized embedding into the retained body tangent space. The pullback of the positive body Hessian is

$$G_O = \iota_O^* K_M \iota_O, \qquad \mathcal{A}_X^{PC} = \langle \Delta r_O, G_O \Delta r_O\rangle, \qquad \Delta r_O = r_{O,1} - r_{O,0}. \tag{23}$$

The expression is coordinate independent and quotient-safe when ι_O descends to the displayed quotient. It has units of action because K_M is the body action Hessian. A8a-OCC and the

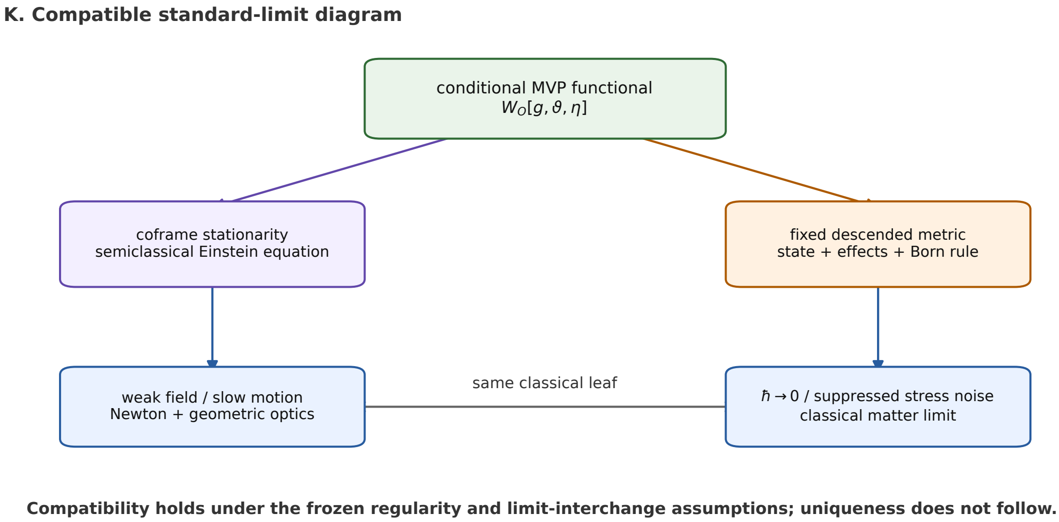

Figure 6: Compatible standard leaves of the frozen completion. Correct limiting behavior is necessary but not sufficient for physical selection or novelty.

support theorem below supply the common amplitude carrier, while A8b supplies the finite global completion of this local metric. Here "Gram amplitude" means an Uhlmann amplitude W satisfying $WW^\dagger = \rho$. Let normalized amplitudes $W_{O,i}$ satisfy $W_{O,i}W_{O,i}^\dagger = \rho_{O,i}$ and $\operatorname{Tr}\rho_{O,i} = 1$. The same inherited action metric is used on their right-unitary quotient, with no independent coefficient:

$$\frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*} = \min_U \|W_{O,1} - W_{O,0}U\|_{\mathrm{HS}}^2, \qquad U^\dagger U = UU^\dagger = I. \tag{24}$$

The amplitudes are taken in a common sufficiently large purification space, so the finite-dimensional optimization can be written over right unitaries. If minimal purification spaces have unequal support, the equivalent partial-isometry formulation is understood; an infinite-dimensional extension would require separate domain and trace-class control.

Lemma 1 (Local metric of the common-Gram minimum lift)**.** *Assume constant rank in a neighborhood of q_0 and choose a smooth local horizontal gauge. Let $W(q)$ be a twice-differentiable normalized amplitude curve through $W(q_0)$, and let $\dot{W}_h$ be its horizontal tangent after minimizing the vertical right-unitary direction. Then*

$$\min_U \|W(q_0 + \delta q) - W(q_0)U\|_{\mathrm{HS}}^2 = \delta q^2 \|\dot{W}_h\|_{\mathrm{HS}}^2 + O(|\delta q|^3). \tag{25}$$

Under A8b, the pullback of this quotient metric is $G_O/\mathcal{A}_*$.

Proof. Write a nearby gauge representative as $W(q_0+\delta q)U(\delta q)^\dagger = W(q_0) + \delta q(\dot{W} - W(q_0)X) + O(\delta q^2)$, with $X^\dagger = -X$. Minimizing the quadratic coefficient over X removes the vertical tangent and leaves $\dot{W}_h$. Squaring the Hilbert–Schmidt norm gives Eq. (25). A8b identifies this horizontal quotient norm, with the common action unit restored, with the body-Hessian pullback in Eq. (23). The last identification is therefore explicit completion structure, not a consequence of Hessian positivity alone. □

A8b is the strongest physical identification in the QOR completion. The lemma proves the local metric of the Uhlmann quotient; it does not prove that the Tau body Hessian owns that metric. Body stationarity, Hessian positivity, endpoint blindness, and readout emergence do not by themselves imply A8b. Section 3.9 derives it only after the additional primary-SHR, pure-realization, recovery, and source-minimal premises have been declared.

Let $\mathcal{A}_* > 0$ be the common elementary action scale and define

$$\epsilon_O^2 = \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*} > 0. \tag{26}$$

Both numerator and denominator have the same units, so ϵ_O is dimensionless. The completion identifies the common action unit with $\hbar$ on the quantum leaf. It then declares a dimensionless quantum-state angle

$$D_Q(\rho_0, \rho_1) = \begin{cases} \arccos |\langle \psi_0 | \psi_1 \rangle|, & \rho_i = |\psi_i\rangle\langle\psi_i|, \\ \arccos \sqrt{F(\rho_0, \rho_1)}, & \text{mixed states,} \end{cases} \tag{27}$$

where

$$F(\rho, \sigma) = \left[\operatorname{Tr} \sqrt{\sqrt{\rho}\, \sigma \sqrt{\rho}} \right]^2$$

is the Uhlmann fidelity. Uhlmann's minimum-lift identity [20] applied to Eq. (24) gives

$$\epsilon_O^2 = 2\left(1 - \sqrt{F(\rho_{O,0}, \rho_{O,1})}\right) = 4 \sin^2 \frac{D_Q(\rho_{O,0}, \rho_{O,1})}{2}. \tag{28}$$

Consequently

$$D_Q(\rho_{O,0}, \rho_{O,1}) = 2 \arcsin \frac{\epsilon_O}{2} \geq \epsilon_O, \qquad 0 < \epsilon_O \leq \sqrt{2}. \tag{29}$$

The final inequality uses $2\sin(x/2) \leq x$ on $0 \leq x \leq \pi/2$. It is therefore a theorem of the common-Gram completion, not a separate distance assumption. Positivity of K_M alone would still be insufficient: the decisive input is the global quotient minimum-lift identity, which already includes all purification shortcuts.

For self-adjoint unitary transport, the Fubini–Study/Bures path obeys

$$D_Q(\rho(0), \rho(\delta t)) \leq \frac{1}{\hbar} \int_0^{\delta t} \Delta H(t)\, dt, \qquad \Delta H(t)^2 = \operatorname{Tr}(\rho H^2) - \operatorname{Tr}(\rho H)^2, \tag{30}$$

with mixed-state extensions expressed through the corresponding quantum Fisher information [16, 2, 19]. Equation (30) is restricted here to the declared self-adjoint unitary protocol. Open or noisy evolution requires its own quantum-Fisher-information path-speed bound and cannot inherit the same ΔH formula without an additional proof. For constant ΔH, combining Eqs. (29) and (30) gives the exact common-Gram bound

$$\boxed{\delta t_{\min,O} \geq \frac{2\hbar}{\Delta H} \arcsin \frac{\epsilon_O}{2} \geq \frac{\hbar \epsilon_O}{\Delta H}.} \tag{31}$$

The inverse-energy-spread form and the Uhlmann chord-to-angle identity are standard, not new Tau laws. The distinct Tau statement is narrower: the source-frozen morphological Gram chord is physically identified with the observer carrier's Uhlmann-amplitude distance. At the same ΔH, two independently reconstructed observer carriers with different source-frozen ϵ_O have different forbidden single-record regions, provided the common-Gram carrier is physically occupied. The numerical prefactor is fixed by the angle convention in Eqs. (27)–(30). An ordering of *observed* onset thresholds follows only if the bound is saturated, or if independently controlled efficiencies establish the same slack above the bound. No detector gain may be fitted to enforce either conclusion.

For an empirical test, "independent reconstruction" therefore has a precise meaning: the carrier class, the two record prototypes, the embedding ι_O, and the pullback metric G_O must be fixed from source-side morphology and calibration data that exclude the timing endpoint. If these objects cannot be operationalized independently, Eq. (31) remains a conditional mathematical discriminator rather than a test-ready numerical prediction.

Here a *single stable record* means that one predeclared protocol run places a pointer variable in one of two disjoint decision regions with single-shot classification error at most $\eta_{\rm rec}$, and that the assignment persists for a predeclared hold time $\tau_{\rm hold}$. No averaging across repeated preparations, tomographic reconstruction, or multi-shot parameter estimation is part of that record event. Those methods may calibrate the carrier or estimate sub-threshold changes, but they test a different operational question.

Finite carrier prototype and uncertainty ledger. A two-level carrier makes the full mathematical chain explicit without claiming physical occupation. The full qubit state carrier is the density- operator space $\mathcal{D}(\mathbb{C}^2)$; $\mathbb{CP}^1$ is only its pure- state projective submanifold. Restrict the prototype records to $\mathcal{C}_O^{\rm pure} = \mathbb{CP}^1$, choose $\rho_{O,0} = |0\rangle\langle 0|$, and set

$$|\psi_{O,1}\rangle = \cos\theta_O|0\rangle + \sin\theta_O|1\rangle, \qquad \rho_{O,1} = |\psi_{O,1}\rangle\langle\psi_{O,1}|.$$

Then $D_Q(\rho_{O,0}, \rho_{O,1}) = |\theta_O|$ for $0 \le |\theta_O| \le \pi/2$. The common-Gram identity gives the exact finite separation

$$\epsilon_O = 2\sin\frac{|\theta_O|}{2}, \qquad |\theta_O| = 2\arcsin\frac{\epsilon_O}{2} \ge \epsilon_O. \tag{32}$$

Thus the two-level example realizes the global theorem rather than assuming a second distance calibration. If a source-declared carrier coordinate q has tangent image $v_O = D\iota_O(\partial_q)$, set $g_O = \langle v_O, K_M v_O\rangle$. In a normal neighborhood,

$$\epsilon_O = |\Delta q|\sqrt{\frac{g_O}{\mathcal{A}_*}} + O(|\Delta q|^2). \tag{33}$$

The local Hessian estimate is not substituted for the exact finite minimum-lift distance outside that neighborhood.

To make "phase-calibrated quasistatic response" reproducible, choose before the timing experiment a closed calibration protocol $\mathcal{P}_{\rm cal}$ and define its unwrapped action phase by $\phi_{\rm cal}(q) = \mathcal{A}_{\rm cal}(q)/\hbar$. Near the source-declared reference q_0, the operational model is

$$\mathcal{A}_{\rm cal}(q) = \mathcal{A}_0 + f_0(q-q_0) + \frac{1}{2}g_O(q-q_0)^2 + O((q-q_0)^3). \tag{34}$$

For a symmetric predeclared step h_q, an endpoint-blind curvature estimator is

$$\frac{\widehat{g}_O}{\hbar} = \frac{\phi_{\rm cal}(q_0+h_q) - 2\phi_{\rm cal}(q_0) + \phi_{\rm cal}(q_0-h_q)}{h_q^2} + O(h_q^2). \tag{35}$$

Equivalently, if $f_q = \partial_q\mathcal{A}_{\rm cal}$ is measured, then $g_O = \partial_q f_q|_{q_0}$. This equality follows rather than being an independent calibration postulate when the protocol is the restriction of the same body action to a source-predeclared curve $M(q)$, with $M(q_0) = M_*$ and $v_O = M'(q_0)$:

$$\frac{d^2}{dq^2}\mathcal{A}_{\rm body}(M(q))\bigg|_{q_0} = K_M(v_O, v_O) + D_M\mathcal{A}_{\rm body}(M_*)[M''(q_0)] = g_O. \tag{36}$$

The final equality uses body stationarity. The neighboring calibration configurations are source-prepared before the timing experiment; this is not terminal backreaction on the already frozen body. Failure of Eq. (36) rejects the proposed common-source carrier realization rather than authorizing a refit.

An endpoint-free calibration must therefore fix the carrier, q, q_0, h_q, phase-unwrapping rule, and $\mathcal{P}_{\rm cal}$ from non-timing device morphology; estimate $g_O/\mathcal{A}_*$ with Eq. (35); determine

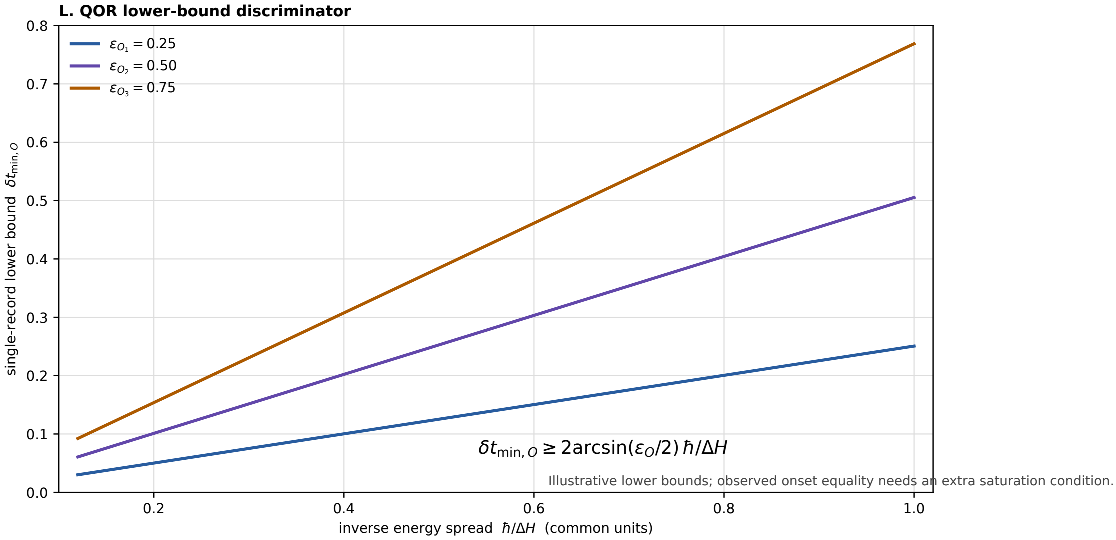

Figure 7: Illustration of Eq. (31). The slopes are not fitted or predicted numerical values. They show the ordering and ratio of *lower-bound* slopes after ϵ_O has been reconstructed independently. Equality of these lines with observed onset thresholds requires an additional saturation or controlled-efficiency condition.

D_Q by state tomography; and determine ΔH by independent energy statistics. The common-Gram identity and Eq. (36) can then be tested before the timing endpoint is opened, avoiding conditional circularity. For a local reconstruction with independent Δq, g_O, and $\mathcal{A}_*$ estimates, first-order uncertainty propagation gives

$$\left(\frac{\sigma_\epsilon}{\epsilon_O}\right)^2 = \left(\frac{\sigma_{\Delta q}}{\Delta q}\right)^2 + \frac{1}{4}\left(\frac{\sigma_g}{g_O}\right)^2 + \frac{1}{4}\left(\frac{\sigma_{\mathcal{A}_*}}{\mathcal{A}_*}\right)^2, \tag{37}$$

or $\sigma_\epsilon^2 = J\Sigma J^T$ with the full calibration covariance. Metrologically $\hbar$ is exact in SI, while its identification with the parent action unit remains a physical assumption of the completion. Two carriers are comparable only under the same Hamiltonian generator and matched ΔH protocol; otherwise the predeclared comparison variable is $\Delta H\,\delta t/\hbar$, not raw time.

This prototype closes the type, normalization, distance bridge, and uncertainty bookkeeping. It also reduces pullback-stiffness reconstruction to the same-body action identity (36). It does not prove that the physical Tau source occupies this common-Gram carrier or supplies a laboratory realization of $\mathcal{C}_O$ and ι_O.

Equation (31) is not a universal time atom and permits continuous parameters. Aggregation may resolve sub-record changes. A test must therefore predeclare the carrier, record criterion, ΔH, aggregation rule, and estimator. Figure 7 shows bounds, not predicted onsets.

Minimal falsification protocol. The test is staged so that a failed carrier is rejected before timing data are used. On one predeclared carrier coordinate and record pair, define

$$\widehat{x}_A = \frac{\widehat{\mathcal{A}_X^{PC}}}{\mathcal{A}_*}, \qquad \widehat{x}_Q = 2\left(1 - \sqrt{\widehat{F}(\rho_{O,0}, \rho_{O,1})}\right). \tag{38}$$

Stage 1 reconstructs $\widehat{x}_A$ from the closed same-action curvature/work protocol and checks Eq. (36), without tomography or endpoint timing. Stage 2 reconstructs $\widehat{x}_Q$ by independent state tomography and tests the Tau-specific identity

$$\boxed{\widehat{x}_A \stackrel{?}{=} \widehat{x}_Q.} \tag{39}$$

For independent estimates, a minimal discrepancy statistic is

$$Z_A = \frac{\widehat{x}_Q - \widehat{x}_A}{\sqrt{\sigma_{x_Q}^2 + \sigma_{x_A}^2}}, \tag{40}$$

with the full declared covariance used when systematic errors are correlated. Defining either estimator from the other makes the test circular. Only a carrier passing both pre-timing tests proceeds to Stage 3, the QOR timing test. Thus failure of the common-Gram realization cannot be hidden by a timing fit.

Choose at least two observer carriers with independently reconstructed $\epsilon_{O_1} \neq \epsilon_{O_2}$, hold the Hamiltonian-spread and record protocol fixed, and estimate the single-record onset without using the timing endpoint in carrier selection. A reproducible stable record at $\delta t < 2\hbar \arcsin(\epsilon_O/2)/\Delta H$, after demonstrating that the one-record, common-Gram minimum-lift, source-freeze, and self-adjoint unitary-protocol conditions hold, rejects the completion. A cross-carrier onset ordering is an additional test only after saturation or equal-slack conditions have been independently justified. Failure of this completion does not by itself reject every possible Tau-style readout architecture.

3.8 Finite source criterion and carrier decision

The two parts of A8 can be reduced to finite source conditions rather than left as an undifferentiated bridge assumption. Let $K_* > 0$ be the body Hessian after the true-null quotient and let

$$R_{\rm rec}^{\rm dedup} : N_*^{\rm phys} \longrightarrow P_{\rm rec}\mathcal{H}_{\rm rec}$$

be the de-duplicated primitive recovery incidence. Define

$$G_*^{\rm rec} = R_{\rm rec}^{\rm dedup} K_*^{-1} (R_{\rm rec}^{\rm dedup})^\dagger, \qquad \rho_*^{\rm rec} = \frac{G_*^{\rm rec}}{\operatorname{Tr} G_*^{\rm rec}}. \tag{41}$$

Proposition 2 (Support-restricted common-amplitude theorem)**.** *Let $K_* > 0$ on a finite retained source tangent space and let $R_{\rm rec}^{\rm dedup} \neq 0$. Set*

$$P_* = \operatorname{supp} G_*^{\rm rec}. \tag{42}$$

Then $\rho_*^{\rm rec}$ is faithful on $P_*\mathcal{H}_{\rm rec}$, whether or not the recovery incidence is onto a larger declared ambient space. If one source-owned CP record instrument $\{\mathcal{I}_i\}$ acts on this occupied support and

$$\rho_i = \frac{\mathcal{I}_i(\rho_*^{\rm rec})}{\operatorname{Tr} \mathcal{I}_i(\rho_*^{\rm rec})} \neq 0, \tag{43}$$

then all ρ_i admit Uhlmann amplitudes in the same Hilbert–Schmidt carrier. A smooth local amplitude bundle exists on every constant-rank stratum.

Proof. For every y,

$$\langle y, G_*^{\rm rec} y \rangle = \left\| K_*^{-1/2} (R_{\rm rec}^{\rm dedup})^\dagger y \right\|^2,$$

so $G_*^{\rm rec} \succeq 0$ and $\ker G_*^{\rm rec} = \ker (R_{\rm rec}^{\rm dedup})^\dagger$. Restriction to $P_*\mathcal{H}_{\rm rec}$ is therefore strictly positive. Functional calculus gives the canonical amplitude $W_* = \sqrt{\rho_*^{\rm rec}}$, and polar decomposition gives every minimal presentation as $W_* U$ up to the usual partial-isometry extension. A CP map has positive output in the same support algebra, so every nonzero conditional state in Eq. (43) has a square-root amplitude in the same Hilbert–Schmidt carrier. Smooth functional calculus and a smooth horizontal gauge give the local bundle at constant rank. □

Thus common-amplitude existence is kinematic once the occupied support and record instrument are supplied. Ambient surjectivity and uniqueness among all stationary parent branches are not needed for this theorem; they would be stronger nature-selection claims. What remains open in A8a-OCC is precisely whether the physical Tau parent occupies this support/instrument packet.

Record-domain and occurrence boundary. No additional apparatus ontology is needed to type one finite record once a positive effect is supplied. For any $0 \preceq E_O \preceq I$, introduce a two-state pointer initialized in $|0\rangle_P$ and set

$$\alpha_O = \arcsin \sqrt{E_O}, \qquad U_{E_O} = \exp(-i\alpha_O \otimes \sigma_y). \tag{44}$$

Projection onto the pointer basis gives

$$K_1 = {}_P\langle 1|U_{E_O}|0\rangle_P = \sqrt{E_O}, \qquad K_0 = {}_P\langle 0|U_{E_O}|0\rangle_P = \sqrt{I - E_O}, \tag{45}$$

up to an irrelevant branch phase. Thus one supplied occurrence produces the canonical square-root CP instrument without a fitted probability gain. Likewise, for any finite orthogonal pointer algebra and supplied record incidences T_j, the positive matching functional

$$\mathcal{A}_{\text{rec}}(r_1, \ldots, r_m \mid p) = \frac{1}{2} \sum_{j=1}^{m} \kappa_j \|r_j - T_j p\|^2, \qquad \kappa_j > 0, \tag{46}$$

has Hessian $\bigoplus_j \kappa_j I > 0$ and the unique structural minimum $r_j = T_j p$. The construction is natural in the finite pointer dimension; it adds neither a Born weight nor a new stiffness law.

Equations (44)– (46) prove availability and stability *conditional on an occurrence*. They do not select whether an observer–record edge is physically occupied. The same static carrier, effect, clock, and conserved current admit a zero-occurrence model and a one-occurrence model. Effect normalization fixes probability conditional on an interrogation, while carrier flux counts occupation rather than interrogations. A finite observer-time persistence interval is also stronger than positivity of the record Hessian: it requires a descended retention generator or direct operational verification. Hence A8a-OCC is reduced to a physical incidence statement, not a missing record-domain type.

A8b has two logically distinct defects. Let $\mathcal{C} : (N_*^{\text{phys}}, K_*) \to (\mathcal{E}_*, h_*)$ be the proposed body-to-amplitude differential and let P_O be the h_*-orthogonal observer-access projector. Define

$$E_{\text{src}} = \mathcal{C}^\dagger h_* \mathcal{C} - K_*, \qquad \Delta_O = \mathcal{C}^\dagger h_* (I - P_O) \mathcal{C} \succeq 0. \tag{47}$$

The actually descended metric is therefore

$$G_O = (P_O \mathcal{C})^\dagger h_* (P_O \mathcal{C}) = K_* + E_{\text{src}} - \Delta_O. \tag{48}$$

The no-loss statement

$$\Delta_O = 0 \quad \Longleftrightarrow \quad \operatorname{im} \mathcal{C} \subseteq \operatorname{im} P_O \tag{49}$$

does not prove $E_{\text{src}} = 0$. A8b follows without a fitted scale only if one and the same parent action supplies the normalized connector identities

$$\boxed{\mathcal{C}^\dagger h_* \mathcal{C} = K_*, \qquad P_O \mathcal{C} = \mathcal{C}.} \tag{50}$$

The first identity is source isometry; the second is complete access only on the selected record tangent, not on the whole parent body. Under both identities, quotienting the right-unitary vertical directions gives the Hilbert–Schmidt/Uhlmann metric with coefficient exactly $K_*/\mathcal{A}_*$.

Multiplying $\mathcal{C}$ by a positive constant would change the first identity and is a different physical source incidence, not a coordinate convention.

Neither common support nor injective conformal descent selects this identity. Indeed, $\mathcal{C}_c = \sqrt{c}\,U$, with U an isometry and $c > 0$, has the same rank, complex structure, conformal cone, and normalized clock-line transport for every c, while its pullback metric changes by c. Consequently A8a-OCC and Hessian positivity do not imply A8b.

The consistency of A8b can nevertheless be exhibited without a fitted coefficient. Write

$$x_O = \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*}. \tag{51}$$

For every $0 \le x_O \le 2$, define

$$r(x_O) = \sqrt{x_O - \frac{x_O^2}{4}}, \qquad \rho_\pm(x_O) = \frac{1}{2}\left(I \pm r(x_O)\sigma_z\right). \tag{52}$$

These are normalized positive qubit states and obey

$$2\left(1 - \sqrt{F(\rho_+, \rho_-)}\right) = x_O. \tag{53}$$

Thus A8b has a normalized mixed-state realization exactly on $0 \le x_O \le 2$; distinct records require $0 < x_O \le 2$. It is also a valid CP observer restriction: the depolarizing map $\mathcal{E}_r(\rho) = r\rho + (1 - r)I/2$ sends an orthogonal qubit pair to Eq. (52) and admits a standard Stinespring isometry V_r.

This proves non-emptiness, not nature-level selection. The current body/Gibbs and CP-descent laws admit every $r \in [0, 1]$ and contain no term that chooses its value. The remaining physical statement is therefore the finite source identity

$$\boxed{V_{\mathrm{phys}} = V_{r(x_O)}, \qquad r(x_O)^2 = x_O - \frac{x_O^2}{4}, \qquad 0 < x_O \le 2.} \tag{54}$$

It must be a second restriction of the same post-body functional that owns x_O. Choosing r from the timing endpoint or measured record overlap would be terminal fitting. Hence A8b is a mathematically closed conditional completion, while its physical source selection remains open.

Proposition 3 (Record occurrence and finite A8b are independent)**.** *Relative to assumptions A1–A9 without an additional law coupling occurrence degree to the amplitude connector, physical record occurrence and the finite A8b identity are logically independent.*

Proof. Fix $x_O \in (0, 2]$ and define two propositions

$$\mathsf{O}: \ n_{\mathrm{rec,O}} \ge 1, \qquad \mathsf{B}: \ 2\big(1 - \sqrt{F(\rho_+, \rho_-)}\big) = x_O, \tag{55}$$

where $n_{\mathrm{rec,O}}$ counts source-owned record incidences in the declared observer support. The depolarizing Stinespring family used above is valid for every $r \in [0, 1]$, and its symmetric output pair has squared Uhlmann chord

$$y(r) = 2\left(1 - \sqrt{1 - r^2}\right). \tag{56}$$

The value $r_*(x_O)$ in Eq. (52) satisfies $y(r_*) = x_O$. Choose any $r_0 \in (0, 1]$ with $r_0 \ne r_*$ and $y(r_0) \ne x_O$. The models $(n_{\mathrm{rec,O}}, r) = (1, r_*)$ and $(1, r_0)$ have the same body ratio, finite pointer type, occurrence degree, and positive record-matching Hessian, but only the first satisfies B. Therefore $\mathsf{O} \not\Rightarrow \mathsf{B}$.

Conversely, hold the carrier geometry fixed at $r = r_*$ and compare $(n_{\mathrm{rec,O}}, r) = (1, r_*)$ with $(0, r_*)$. The finite metric identity holds in both admissible carrier completions, while only the first contains a physical record event. Thus $\mathsf{B} \not\Rightarrow \mathsf{O}$. Repeating the zero/one occurrence choice at r_0 realizes all four truth assignments of (O, B), proving independence. The zero-occurrence A8b case is a structural law on an available carrier, not a measured record. $\square$

The proposition is relative to the current source axioms, not a prohibition on deeper unification. A richer post-body source law may select occurrence and the connector jointly. In the present completion, however, a realized QOR trial requires the conjunction

$$\boxed{n_{\mathrm{rec,O}} \geq 1 \quad \wedge \quad x_A = x_Q \quad \wedge \quad \mathsf{H}_{\mathrm{hold}}.} \tag{57}$$

Here $\mathsf{H}_{\mathrm{hold}}$ denotes the independently predeclared single-record retention condition. Occupation supplies an event, A8b supplies its finite morphology/Bures calibration, and the hold condition makes the event operationally readable; none substitutes for another.

3.9 Finite source closure: foundation-level summary

This subsection records only the source-level consequences needed by the foundation argument. The detailed complete-edge, holonomy and irreversible CP analysis and the detailed finite-signature, marginal and incidence proofs are technically separable. Their common message is that internal representation closure can be proved under explicit finite source data, while physical selection of those data remains open.

Blockwise spectral-to-amplitude transfer. On an occupied recovery decomposition, one source-owned connector with a single inherited action unit can transport orthogonal spectral blocks to orthogonal Uhlmann-amplitude blocks. Local metric agreement is insufficient for finite equality; the finite response must additionally be source-owned and extensive on its admitted direct-sum domain.

Sufficient internal realization of finite A8b. Inside the declared enriched special dagger-Frobenius packet, a complete source-owned record edge is faithfully trace preserving. Its standard-form lift is isometric, and primitive-defect extensivity forces

$$\frac{\mathcal{X}_s(e)}{\mathcal{A}_*} = d^2_{U,\mathrm{chord}}(\rho_u, \rho_v). \tag{58}$$

This is a sufficient internal realization theorem, not a proof that physical record edges are reversible or occupied.

Pathwise A8b and the holonomy boundary. Composable complete edges preserve Eq. (58) pathwise. Equal endpoints may be collapsed only when relative holonomy is Uhlmann-gauge or null under a source-frozen claim-complete descriptor. Descriptor-visible holonomy remains physical path information. A generic irreversible CPTP relation instead requires a certified recovery on its frozen occupied domain; commuting records alone cannot identify reversibility.

Proposition 4 (Primary-SHR recovery differential generates affiliation)**.** *Within the declared source-minimal primary-SHR realization, a source-owned nonzero nilpotent recovery differential and its Hodge–Dirac operator are affiliated with the finite recovery algebra. With the common-action amplitude connector this is sufficient for finite A8b.*

Proof sketch. The reconstructed seed boundary descends to the physical quotient, its differential squares to zero, and the differential and adjoint belong to the finite recovery algebra. Their self-adjoint sum is therefore affiliated. The statement remains conditional on the declared source-minimal realization and connector. □

Proposition 5 (Universal-seed rich-body recovery)**.** *If every occupied base–seed response stabilized before descent is included in one complete body, the body action is basic on the true-null quotient, and the universal seed owns the complete ordering presentation, then the recovery* *differential and affiliated Hodge–Dirac construction extend to the full body without requiring minimum body dimension.*

Proof sketch. The full realization defect vanishes under action basicness. Complete seed-owned order confines any chain mismatch to decoder-vertical redundancy; true-null descent removes only terminal-invisible directions. A visible decoder-null ordering component would falsify the premise and require a richer seed or an order-generating base law. □

3.9.1 Exact open step: physical selection

The internal results do not close the physical arrow:

$$B_\tau^{\mathrm{phys}} + s_U \quad \not\Longrightarrow_{\text{current derivation}} \quad \Gamma_{\mathrm{MVP}} + \text{state/effect completion} + \text{QOR carrier}. \tag{59}$$

The open packet includes the universal seed, complete body/order premises, occupied record support, same-source identity $x_A = x_Q$, coherent joint source map and observer hold. No downstream consistency theorem proves that Nature occupies these inputs.

3.9.2 Finite source-signature result

The finite source layer has one compact conclusion. For a finite source algebra $\mathcal{A}_s$, faithful state ω_s, and complete typed role packet, let Σ_s denote the multiplication, involution, state and distinguished-incidence data. Then:

Proposition 6 (Finite source-signature representation uniqueness and joint-incidence boundary)**.** *Two minimal cyclic finite realizations of the same complete faithful typed signature are unitarily equivalent by the unique typed unitary mapping one cyclic vector to the other. This proves representation uniqueness after complete source ownership; it does not supply that ownership. Separate sector marginals do not determine the joint source, and even all proper marginals in the four-qubit Pauli control leave the 81-dimensional weight-four subspace unresolved. Conversely, one source-owned onto joint incidence with positive stiffness produces a faithful joint Gram state.*

Proposition 7 (Sector marginals do not select the joint source)**.** *Faithful marginal states do not generally determine their joint source state.*

Proposition 8 (All proper marginals remain incomplete)**.** *In the four-qubit Pauli control, all proper marginals span 174 of 255 traceless directions and leave 81 genuine four-body directions unresolved.*

Proposition 9 (One onto joint incidence is sufficient)**.** *If $K_s \succ 0$ and the source-owned incidence $R_J : \mathcal{E}_J \to \mathcal{J}_s$ is onto, then the normalized Gram state $\rho_J \propto R_J K_s^{-1} R_J^\dagger$ is faithful and fixes every finite mixed moment.*

The proofs are finite: GNS gives typed cyclic uniqueness, the family $(I_{16} + \epsilon\sigma_z^{\otimes 4})/16$ proves the proper-marginal no-go, and surjectivity makes $R_J^\dagger$ injective. The once-owned primitive roles therefore compress to one coherent physical target

$$\Pi_{\mathrm{phys}} : \mathcal{A}_{\mathrm{prim}} \longrightarrow \mathcal{B}(\mathcal{H}_{\mathrm{post}}), \tag{60}$$

whose kernel selects the relation ideal and whose quotient image selects the representation. Its nature-level occupation remains open. The full proofs, scope conditions and representation-specific dimension audit are technically separable from this foundation statement.

3.9.3 Source-generated records and atlas summary

Proposition 10 (Dagger unfolding supplies the record-copy map)**.** *Inside the enriched special dagger-Frobenius completion, the adjoint unfolding map* $\Delta = \mu^\dagger$ *with* $\mu\Delta = I$ *supplies an isometric record-copy mechanism without a free gain.*

Proof sketch. Specialness gives $\Delta^\dagger\Delta = I$. The resulting copy map is therefore normalized in the inherited source metric. The unchanged narrow parent does not contain or select this two-leg carrier. □

At fixed occupied roots and resolution, closure under the frozen incidences gives an inclusion-minimal observer atlas. GNS, edge and atlas constructions organize source-owned roles; they create none. Missing occurrence, hold, finite action or microscopic ownership remains a physical blocker.

3.10 Dimensional and claim-status audit

The principal quantities used by the completion have the following roles.

Quantity	Dimensions	Status
M_*, K_M	branch-dependent body coordinate and action Hessian	assumed stabilized; positive after true-null quotient
G_{SHR}	$L^3M^{-1}T^{-2}$ in SI	not numerically predicted
Λ_{ren}	L^{-2}	renormalized infrared parameter; not predicted
$\mathcal{A}_X^{PC}, \mathcal{A}_*$	action	common units required
$x_O = \mathcal{A}_X^{PC}/\mathcal{A}_*$	dimensionless	body-action record ratio; A8b requires $0 < x_O \leq 2$ for distinct records
D_Q	dimensionless angle	adopted Fubini–Study/Bures distinguishability metric
ϵ_O	dimensionless	source-frozen Gram chord; $D_Q = 2\arcsin(\epsilon_O/2)$
ΔH	energy	protocol-frozen energy spread
$\delta t_{\mathrm{min},O}$	time	conditional single-record lower bound
$\eta_{\mathrm{rec}}, \tau_{\mathrm{hold}}$	dimensionless probability, time	predeclared single-shot error ceiling and record-persistence interval

The resulting claim ledger is finite:

Statement	Status in this paper
Support-restricted common Uhlmann-amplitude bundle	derived once the occupied support and CP record instrument are supplied; physical occupation open
A8b mixed-state realization and $0 < x_O \leq 2$ range	derived consistency result; finite A8b follows from affiliation plus the explicit source-owned block connector, while nature-level occupation remains open
Complete-record-edge finite A8b	derived inside the enriched special dagger-Frobenius completion: the edge map is faithfully trace preserving and primitive extensivity fixes the exact Trace–Gram law; arbitrary-state and nature/laboratory occupation remain open
Pathwise finite A8b	derived on the occupied connected record groupoid; endpoint-only descent holds only for descriptor-null or Uhlmann-gauge holonomy, while visible holonomy remains typed path information

Statement	Status in this paper (continued)
Complete-edge existence	not derived; the theorem characterizes a sufficient reversible source relation, while irreversible CP/recovery-correctable generalizations remain open
Reversible versus irreversible occupation	provably indistinguishable on a commuting record algebra; a coherent off-diagonal witness or independently verified recovery map is necessary
Frozen holonomy null policy	defined by the complete source-frozen endpoint descriptor; central or path automorphisms are not null by default
Exhaustive rich-body extension	complete stabilized base response has zero complete-body realization defect; universal-seed chain descent preserves the affiliated recovery construction without minimum body dimension
Record occurrence versus finite A8b	logically independent relative to the current source axioms; a physical QOR trial requires occurrence, A8b, and operational hold jointly
Finite source signature and joint incidence	typed cyclic representation is unique after the packet is supplied; marginals remain incomplete, while one onto incidence is sufficient for a faithful joint Gram state
Dagger-Frobenius record unfolding	conditional theorem with no free record gain; enriched two-leg base ownership open
Primitive occupied observer atlas	unique at fixed record roots and resolution; physical occupation and observer count open
Once-counted effective and quadratic ledger	derived once microscopic ownership and renormalization prescription are supplied; full microscopic Tau action open
Semiclassical Einstein/optics leaf	conditional standard-limit recovery
Trace-form Born leaf on fixed or classical-dynamic geometry	conditional on the state/effect valuation and covariant-transport assumptions
Born structure on a quantized metric	compatible only on a supplied constrained joint geometry–matter algebra; current construction open
Compatibility of classical limits	conditional consistency result
Pre-timing identity $x_A = x_Q$	Tau-specific conditional discriminator; physical source value and carrier occupation open
QOR inequality (31)	standard Bures-angle quantum speed limit after A8b; downstream consistency/falsification stage, not an extra Tau delay
Unique selection from the physical base and seed	exact finite criterion derived; current complete source tuple and cross-sector ownership not proved
Numerical constants, UV completion, full Standard Model	open
Empirical support for the Tau-specific common-source identification	open

4 Formal safeguards, postulates, and finite controls

The parent/readout notation is scientifically empty if it can be reduced to $\mathfrak{R}_\tau = \mathcal{O}$, $\rho_\tau(s) = O$, and $U = \mathrm{id}$. A promoted branch must therefore impose, before endpoint inspection, at least one nontrivial source relation, quotient-safe null, shared-source constraint, known-limit recovery, wrong-family exclusion, or other consequence that could fail. Known-limit recovery alone is not enough because an existing endpoint theory can always be harmlessly relabelled.

For readout U_i, let N_i be its declared null relation. Quotient safety means

$$rN_ir' \Longrightarrow U_i(r) = U_i(r'), \qquad U_i = \bar{U}_i \circ q_i, \quad q_i : \mathfrak{R}_\tau \to \mathfrak{R}_\tau/N_i. \tag{61}$$

The converse need not hold: additional identifications constitute readout loss. A stable observed

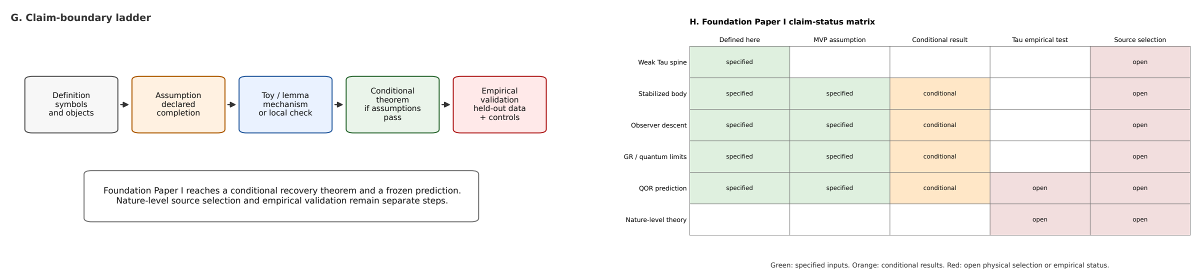

Figure 8: The claim ladder and status matrix. Definitions, postulates, conditional theorems, toys, numerical evidence and validation are distinct; the present paper reaches conditional standard leaves and a discriminator, not nature selection or empirical validation.

object class may be represented by a basin $\mathcal{B}_O = \{r : R_{4D}(r) \simeq O\}$; this is a persistence scaffold, not a particle-stability or decay theorem. Protected source identity must not be collapsed unless that loss was declared as null structure.

Endpoint blindness operationalizes the same rule. Before scoring a target, one must freeze the source or proxy, readout family, null/quotient and overlap policies, forbidden endpoint fields, controls, failure conditions, and a dated freeze artifact. A residual used to choose the source class that later "explains" it is endpoint leakage, not a Tau prediction.

The seven unproved foundation postulates are: (P1) an atemporal Tau substrate; (P2) endpoint-blind seed/base response stabilizing a body; (P3) endpoint-blind source and kernel selection; (P4) complete physical descriptions as typed readouts, not residual-only corrections; (P5) effective variation only after a stable readout ordering or clock is available; (P6) a hidden defect only when relift fails modulo the declared null; and (P7) observational tensions as possible readout/closure defects only after controls and held-out tests. These postulates define the candidate framework; the MVP realizes one conditional branch but does not derive the postulates from the physical base–seed law.

A finite control makes the quotient discipline explicit. Let $r = (a, b, c) \in \mathbb{R}^3$,

$$U_A(r) = (a, b), \quad U_B(r) = (a, c), \qquad J_A(x, y) = (x, y, 0), \quad J_B(x, z) = (x, 0, z).$$

Then $h_A = (0, 0, -c)$ and $h_B = (0, -b, 0)$: hiddenness is readout-relative. Replacing U_A after inspection by $U_A^{\mathrm{bad}}(a, b, c) = (a, b + \alpha c)$ while retaining the old null is inconsistent. Likewise, a source rule $c = f_{\mathrm{src}}(a, b; \theta)$ is admissible only if f_{src} and θ were frozen upstream.

5 Atemporal body and effective readout-time

Let $r = \rho_\tau(s)$ be atemporal and suppose $U_i(r) = o_i(t_i)$. The derivative do_i/dt_i is meaningful inside that readout without being a parent-time derivative. Another readout of the same response may return a static constraint. There is no contradiction: time is the stable ordering used by one terminal description, not an external clock that evolves the parent.

For a finite witness, take $r = (a_0, \ldots, a_n, c)$, $U_{\mathrm{seq}}(r) = (a_0, \ldots, a_n)$, and $U_{\mathrm{sum}}(r) = \sum_k a_k$. The first supports the ordered difference $\Delta a_k = a_{k+1} - a_k$; the second loses that order. The parent record has not evolved. The readout has exposed an ordering.

The morphological body M_τ is the structured source object conditioning these readouts, not a visible galaxy shape or catalogue label:

$$O_i = U_i^{M_\tau}(\rho_\tau(s)). \tag{62}$$

Images, rotation curves, lensing maps, cosmic-web descriptors and measurement contexts are projected handles only. They can underdetermine global incidence, order, path geometry, closure and null structure. Body selection, kernel family and source overlaps must therefore be

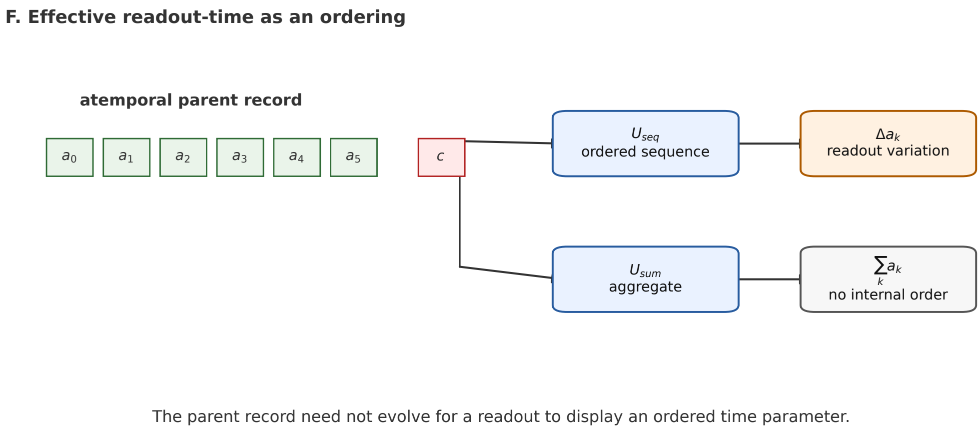

Figure 9: Effective readout-time in a finite toy. A sequence readout exposes ordering and finite differences; an aggregate readout of the same parent record does not.

frozen before endpoint scoring, and two terminals using the same source coordinate may not be counted independently without an upstream split.

Observer time is correspondingly sector-typed:

$$t^S_{\text{obs}} = U^S_{\text{time}}(\rho_\mathcal{T}(s), M_\mathcal{T}, \mathcal{O}_S, \gamma_S). \tag{63}$$

The MVP adds an oriented clock descent and the conditional QOR law, but does not prove occupation of that clock carrier. Proper-time recovery and any nonrelativistic observer correction require a noncircular branch test. Future-labelled components of the parent record do not imply backward causation; the temporal order belongs to the descended readout.

6 Primitive readout families

The weak quantum notation $q = U_q^{M_\mathcal{T}}(\rho_\mathcal{T}(s))$ does not imply Hilbert space, interference or the Born rule. The MVP obtains trace form only after the positive state/effect assumptions A6; information loss or finite capacity alone gives coarse graining, not complex quantum mechanics [12, 6, 17, 3]. Quantum stress and semiclassical gravity share the common functional, but metric quantization, interacting UV QFT and a complete measurement ontology remain open.

For gravity, an obstruction is meaningful only as $\omega_g(r) = 0 \Leftrightarrow J_g U_g(r) N_g r$. A chart difference $J_g U_g(r) - r$ is secondary, and an empirical residual may not select the body. Dark-sector inference can mix gravitating content, baryonic remnants, modified response, morphology, path, observer and calibration effects. The conditional Einstein leaf does not choose among them or explain dark matter or dark energy.

A cosmological readout may be typed as

$$O_{\text{cosmo}} = U_{\text{cosmo}}(\rho_\mathcal{T}(s), M^U_\mathcal{T}, \gamma_{\text{path}}, \mathcal{O}_{\text{obs}}).$$

This warns against compressing expansion, distance, growth, lensing and CMB tensions into one unconstrained correction. A physical branch requires multi-probe covariance, source-native descriptors and controlled ΛCDM limits. Neither Λ_{ren} nor its numerical value is derived here.

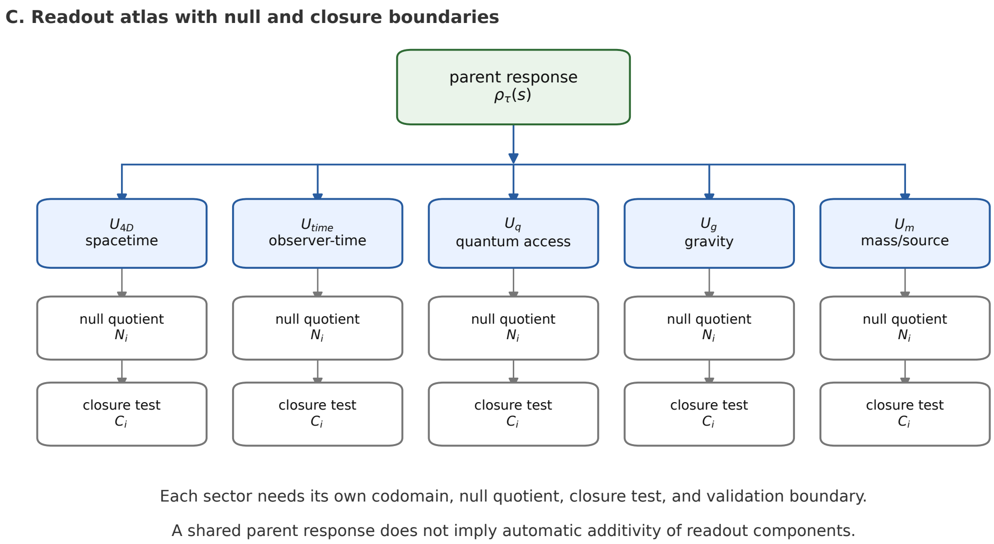

Figure 10: A shared parent response can support several typed readout families. Their codomains, nulls, closure tests and ownership must be audited before they are counted as independent information.

7 Physical positioning and adjacent programs

Tau Core is neither anti-MOND nor anti-dark-matter. MOND-like laws may be effective limits, while lensing, CMB, BAO, clusters and growth still require joint source-frozen tests; this paper does not claim that collisionless matter is illusory. Relational clocks [18, 7], shape dynamics [11], semiclassical gravity [15], locally covariant QFT [4, 13, 14] and holographic error correction [1] are adjacent programs, not evidence for Tau ontology. The inverse-energy QOR form is a standard speed-limit structure [16, 2, 19]; only the independently reconstructed same-source body–Uhlmann identity is proposed as Tau-specific.

8 Falsifiability, objections, and next work

Tau Core becomes physics only through bounded source-first tasks. A branch must declare a noncircular response law and source-realizable body, descend through a quotient-safe readout, recover the appropriate known limit, and freeze at least one consequence that could fail before inspecting the endpoint. Negative results must demote a branch rather than authorize a new terminal or retuned kernel.

The principal failure modes are: no source-realizable parent object; a readout that is not well defined on its declared quotient; failure of the relevant GR, Newtonian, Born or cosmological limit; endpoint leakage; and rescue by an unconstrained post-hoc terminal. The corresponding status labels are *toy-only*, *phenomenological-only*, *blocked* and *rejected*. Toy-only means that a construction is internally consistent but not physically selected; phenomenological-only means that an effective fit exists without a parent derivation; blocked means that a named missing theorem or datum prevents promotion; rejected means that a frozen prediction, quotient condition, or known limit has failed. In particular, a coordinate declared null cannot later be used as an endpoint-selected correction without failing quotient safety.

The strongest reviewer objections are therefore legitimate tests, not matters of terminology. Without source-frozen constraints the parent is only a relabeling; without a physical selection law it is unobservable scaffolding; without preserved negative results the framework could explain anything. The displayed Einstein and Born leaves are conditional standard recovery, not first-principles derivations from the weak Tau spine. The QOR inverse-energy dependence is standard; only the independently reconstructed same-source identity $x_A = x_Q$ is Tau-specific. The once-counted ledger removes formal double counting only after microscopic ownership is supplied. Quantized or strongly fluctuating geometry and extreme semiclassical regimes remain outside the current theorem.

Structural novelty must take the form

$$\mathrm{Struct}_\tau(s, M_\tau) = 0 \Longrightarrow \mathrm{Constraint}(U_i^{M_\tau}, U_j^{M_\tau}, N_i, N_j),$$

or an upstream exclusion of a readout pattern. Merely adding a residual term to MOND, ΛCDM, GR or quantum theory does not qualify. At minimum a promoted branch must provide source declaration, body-first order, common ownership, readout codomain, null and relift policies, quotient safety, known limit, dimensional audit, a new discriminator and an explicit failure rule.

The next decisive task is the first open arrow

$$B_\tau^{\mathrm{phys}} + s_U \longrightarrow \Gamma_{\mathrm{MVP}} + \text{state/effect completion} + \text{observer carrier}.$$

It requires physical selection of the coherent Π_{phys} packet, occupied record support, the same-action A8b connector, universal-seed order compatibility, microscopic once-counting and record hold. A laboratory route must independently reconstruct body action, state distance, coherent/recovery evidence, energy spread and timing on one frozen carrier. No platform is claimed to have completed that certificate. Superconducting qubits, trapped-ion or spin systems, metastable atomic levels and photon polarization are candidate carriers only, not asserted Tau realizations. Likewise, cross-readout residuals are diagnostics only after matched null, covariance and source-freeze controls; they are not by themselves validation statistics.

9 What is not proved

For clarity, we list the main non-results.

- The paper does not prove that the Tau substrate, universal seed, or complete universe-scale morphological body exists physically.
- It does not derive the adopted enriched body action and completion uniquely from the physical base–seed source.
- It does not derive the microscopic ownership partition or full nonlinear action whose integration yields the once-counted effective functional in Eq. (12).
- It does not predict G, Λ, $\hbar$, gauge couplings, particle masses, or other constants numerically from parent data.
- Conditional Einstein and Born recovery is not an unconditional derivation of GR or quantum mechanics from the minimal Tau base.
- It does not provide ultraviolet quantum gravity, nonperturbative interacting QFT, a complete measurement ontology, or a metric-quantization theorem.
- It does not derive the full Standard Model, its spectrum, generations, mixing, or neutrino sector.
- The QOR inequality is not a universal time atom or an extra delay. The support theorem and mixed-state realization prove that the common-Gram completion is mathematically nonempty, but the physical parent has not been shown to occupy the record instrument or select $V_{\mathrm{phys}} = V_{r(x_O)}$. No laboratory carrier has validated the same-source identity $x_A = x_Q$.

- The dagger-Frobenius unfolding theorem supplies a normalized record-copy mechanism only inside an enriched two-leg base completion. The present narrow parent action neither contains nor selects that carrier. Once an occupied complete record edge of that enriched packet is supplied, finite A8b is derived on its complete path orbit; the paper does not prove that Nature or a laboratory carrier occupies the orbit or that it covers arbitrary quantum states.
- The complete-edge theorem does not cover generic irreversible CP, decohering or rank-reducing record processes. Exact A8b for such processes would require a separately proved recovery-correctable or equivalent nonisomorphic extension.
- The finite joint-state and colimit reductions do not prove that Nature occupies the coherent source homomorphism Π_{phys}; all proper marginals remain insufficient.
- It does not eliminate or explain dark matter or dark energy.
- No empirical result presented in this manuscript validates the Tau ontology.

These limitations are not footnotes. They are part of the claim. A foundation paper that hides its blockers is not useful. The goal is to make the blockers visible enough that later work can attack them one at a time.

10 Conclusion

Tau Core is presented as an atemporal morphological readout framework with one frozen conditional completion. A stabilized body is solved before terminal variation, and one once-counted functional supplies compatible conditional semiclassical-Einstein/optics and Schrödinger–Born leaves. This is an auditable architecture, not a theorem that the physical base–seed source selects it.

The proposed finite discriminator is the independently reconstructed same-source identity

$$x_A = \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*} = x_Q = d^2_{U,\text{chord}}(\rho_0, \rho_1),$$

which turns the standard quantum speed limit into the QOR bound. The novelty claim concerns this body–Uhlmann bridge, not inverse-energy timing and not a universal time atom.

Inside the declared enriched packet, a complete source-owned record edge and primitive extensivity are sufficient for exact finite A8b. Pathwise descent retains descriptor-visible holonomy, and commuting records do not identify reversibility. The strengthened FOC-7 certificate therefore requires a coherent witness or certified recovery in addition to independent action, tomography and timing measurements.

The decisive blocker remains nature-level selection and occupation of the common source packet. A deeper source theorem or a successful source-frozen FOC-7 test could promote this completion; a reliable violation would reject it. Adding unconstrained terminals after such a failure would abandon, not repair, the minimum-viable theory tested here.

A Mixed-state distance and QFI conventions

To avoid the common squared-versus-root fidelity ambiguity, this manuscript uses the *squared* Uhlmann fidelity

$$F(\rho, \sigma) = \left[\mathrm{Tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}\right]^2, \qquad 0 \le F \le 1. \tag{64}$$

The root fidelity is $f(\rho, \sigma) = \sqrt{F(\rho, \sigma)}$. The Bures angle used in Eq. (27) is therefore

$$D_B(\rho, \sigma) = \arccos f(\rho, \sigma) = \arccos\sqrt{F(\rho, \sigma)}. \tag{65}$$

For pure states, $F = |\langle\psi|\varphi\rangle|^2$, so $D_B = \arccos|\langle\psi|\varphi\rangle|$, the Fubini–Study angle.

For a differentiable one-parameter mixed-state path, the infinitesimal Bures line element and quantum Fisher information $\mathcal{F}_Q$ satisfy

$$ds_B^2 = \frac{1}{4}\mathcal{F}_Q(t)\,dt^2. \tag{66}$$

For unitary evolution generated by a self-adjoint H,

$$\mathcal{F}_Q(t) \le \frac{4\Delta H(t)^2}{\hbar^2}, \qquad \frac{ds_B}{dt} \le \frac{\Delta H(t)}{\hbar}, \tag{67}$$

which integrates to Eq. (30) [19]. A nonunitary channel must instead use its actual QFI path speed; Eq. (67) is not assumed for it.

The common-Gram completion fixes the notation

$$x_O = \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*} = \epsilon_O^2 = 2(1-\sqrt{F}), \qquad D_Q = 2\arcsin(\epsilon_O/2).$$

Hence the linear inequality $D_Q \ge \epsilon_O$ is a conservative corollary, with equality only in the zero-separation limit. The exact state-distance quantum speed limit is the first inequality in Eq. (31).

References

[1] Ahmed Almheiri, Xi Dong, and Daniel Harlow. Bulk locality and quantum error correction in AdS/CFT. *Journal of High Energy Physics*, 2015(4):163, 2015.

[2] Jeeva Anandan and Yakir Aharonov. Geometry of quantum evolution. *Physical Review Letters*, 65:1697–1700, 1990.

[3] Howard Barnum and Alexander Wilce. Post-classical probability theory. *arXiv:1205.3833*, 2014.

[4] Romeo Brunetti, Klaus Fredenhagen, and Rainer Verch. The generally covariant locality principle—a new paradigm for local quantum field theory. *Communications in Mathematical Physics*, 237:31–68, 2003.

[5] Paul Busch. Quantum states and generalized observables: A simple proof of gleason's theorem. *Physical Review Letters*, 91:120403, 2003.

[6] Giulio Chiribella, Giacomo Mauro D'Ariano, and Paolo Perinotti. Informational derivation of quantum theory. *Physical Review A*, 84:012311, 2011.

[7] Alain Connes and Carlo Rovelli. Von neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories. *Classical and Quantum Gravity*, 11(12):2899–2917, 1994.

[8] David Deutsch. Constructor theory. *Synthese*, 190:4331–4359, 2013.

[9] Bryce S. DeWitt. Quantum theory of gravity. i. the canonical theory. *Physical Review*, 160:1113–1148, 1967.

[10] Andrew M. Gleason. Measures on the closed subspaces of a hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.

[11] Henrique Gomes, Sean Gryb, and Tim Koslowski. Einstein gravity as a 3d conformally invariant theory. *arXiv:1010.2481*, 2011.

[12] Lucien Hardy. Quantum theory from five reasonable axioms. *arXiv:quant-ph/0101012*, 2001.

[13] Stefan Hollands and Robert M. Wald. Local wick polynomials and time ordered products of quantum fields in curved spacetime. *Communications in Mathematical Physics*, 223:289–326, 2001.

[14] Stefan Hollands and Robert M. Wald. Existence of local covariant time ordered products of quantum fields in curved spacetime. *Communications in Mathematical Physics*, 231:309–345, 2002.

[15] Bei-Lok B. Hu and Enric Verdaguer. Stochastic gravity: Theory and applications. *Living Reviews in Relativity*, 11:3, 2008.

[16] Leonid Mandelstam and Igor Tamm. The uncertainty relation between energy and time in non-relativistic quantum mechanics. *Journal of Physics (USSR)*, 9:249–254, 1945.

[17] Lluis Masanes and Markus P. Mueller. A derivation of quantum theory from physical requirements. *New Journal of Physics*, 13:063001, 2011.

[18] Don N. Page and William K. Wootters. Evolution without evolution: Dynamics described by stationary observables. *Physical Review D*, 27:2885–2892, 1983.

[19] M. M. Taddei, B. M. Escher, L. Davidovich, and R. L. de Matos Filho. Quantum speed limit for physical processes. *Physical Review Letters*, 110:050402, 2013.

[20] Armin Uhlmann. The "transition probability" in the state space of a *-algebra. *Reports on Mathematical Physics*, 9(2):273–279, 1976.

[21] Robert M. Wald. The back reaction effect in particle creation in curved spacetime. *Communications in Mathematical Physics*, 54:1–19, 1977.